A Subtree L-Summand Partial Answer for Super Daugavet Points in Free Spaces over R-Trees

Automatic Math Discovery Run `fa_banach_001`

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Abstract

Abrahamsen, Aliaga, Lima, Martiny, Perreau, Prochazka, and Veeorg ask whether arbitrary Daugavet-points, respectively Δ-points, in Lipschitz-free spaces over subsets of $\mathbb{R}$-trees must be super Daugavet-points, respectively super Δ-points. The source paper proves this for molecules and for convex combinations or convex series of molecules.

This packet proves a further positive subcase. If a point of $\mathcal{F}(M)$ lies in the free space over a closed metric-convex subtree $A \subset M$, then it is a super Daugavet-point of the ambient space $\mathcal{F}(M)$. The key step is that the nearest-point retraction onto A linearizes to an L-projection $\mathcal{F}(M) \to \mathcal{F}(A)$. Since $\mathcal{F}(A)$ is an atomless L_1 space, all of its unit vectors are super Daugavet-points, and this transfers through the resulting ℓ_1-sum decomposition.

The arbitrary-subset problem remains open here: the argument does not control points whose Godard L_1 representation is tied to non-convex subsets or to missing-branch constraints.

1 Source question

The source paper is:

> T. A. Abrahamsen, R. J. Aliaga, V. Lima, A. Martiny, Y. Perreau, A. Prochazka, and T. Veeorg, *A relative version of Daugavet-points and the Daugavet property*, arXiv:2306.05536.

In Section 4.1 the authors prove that, for complete subsets of $\mathbb{R}$-trees, Daugavet-points and Δ-points coincide. They also prove the super upgrade for molecules and for convex combinations or convex series of molecules. Immediately afterward they write:

> Let us finally point out that we do not know whether arbitrary Daugavet-points in Lipschitz-free spaces over subsets of $\mathbb{R}$-trees need always be super Daugavet-points, or whether arbitrary Δ-points need always be super Δ-points.

Thus the remaining question is the arbitrary unit-vector case.

2 Statement

Let T be an $\mathbb{R}$-tree. A subset $A \subset T$ is called *metric-convex* if $[a, b] \subset A$ for all $a, b \in A$.

Theorem 2.1. *Let T be a complete $\mathbb{R}$-tree, let $M \subset T$, and let $A \subset M$ be a non-degenerate closed metric-convex subset of T. Then $\mathcal{F}(A)$ is an L-summand in $\mathcal{F}(M)$.*

Corollary 2.2. *Under the hypotheses of Theorem 2.1, every $\mu \in S_{\mathcal{F}(A)}$ is a super Daugavet-point of $\mathcal{F}(M)$. Consequently, if M itself is a non-degenerate complete metric-convex subset of an $\mathbb{R}$-tree, then every element of $S_{\mathcal{F}(M)}$ is a super Daugavet-point.*

3 Proof intuition

For a segment $[x, y] \subset M$, the source paper proves that the canonical retraction $Y_{xy} : M \to [x, y]$ induces an L-projection $\mathcal{F}(M) \to \mathcal{F}([x, y])$. The same proof is not special to intervals. In an $\mathbb{R}$-tree, the nearest-point projection onto any closed convex subtree A has the same geodesic additivity property: if two projected points differ, the geodesic from p to q passes from p to $\pi_A p$, then through A, then from $\pi_A q$ to q. This lets one paste two norming Lipschitz functions exactly as in the segment proof and obtain

$$\|\mu\| = \|\widehat{\pi_A}\mu\| + \|\mu - \widehat{\pi_A}\mu\| .$$

After that the problem is no longer free-space-specific: $\mathcal{F}(A)$ is the free space over an $\mathbb{R}$-tree, hence an atomless L_1 space by Godard, so all its unit vectors are super Daugavet-points. Super Daugavet-points pass from an L-summand to the ambient ℓ_1-sum.

4 Proof

We use the standard base-point independence of Lipschitz-free spaces and choose the base point 0 in A.

Lemma 4.1. *Let T be a complete $\mathbb{R}$-tree and let $A \subset T$ be nonempty, closed, and metric-convex. For every $p \in T$ there is a unique nearest point*

$$\pi(p) \in A, \qquad d(p, \pi(p)) = \operatorname{dist}(p, A).$$

The map $\pi : T \to A$ is a 1-Lipschitz retraction. Moreover, if $\pi(p) \neq \pi(q)$, then

$$d(p, q) = d(p, \pi(p)) + d(\pi(p), \pi(q)) + d(\pi(q), q).$$

Proof. These are the standard projection properties of closed convex subsets of complete $\mathbb{R}$-trees. Indeed, the unique arc from p to A meets A first at $\pi(p)$; uniqueness follows from convexity, since two different nearest points would force the segment between them to lie in A and contain a closer point. If $\pi(p) \neq \pi(q)$, the segment $[\pi(p), \pi(q)]$ lies in A. The geodesic from p to q must enter A at $\pi(p)$ and leave it at $\pi(q)$, otherwise one of the two projections would not be nearest. The displayed distance identity follows from additivity of length along this concatenated geodesic. The same identity, with the degenerate case allowed, also gives that π is 1-Lipschitz. $\square$

Proof of Theorem 2.1. Let $P = \widehat{\pi|_M} : \mathcal{F}(M) \to \mathcal{F}(A)$ be the linearization of the nearest-point retraction. It is a norm-one projection. We show that it is an L-projection.

It is enough to prove

$$\|\mu\| = \|P\mu\| + \|\mu - P\mu\|$$

for finitely supported $\mu \in \mathcal{F}(M)$, since such elements are dense in $\mathcal{F}(M)$. Fix such a μ. Choose $f, g \in B_{\mathrm{Lip}_0(M)}$ with

$$\langle P\mu, f\rangle = \|P\mu\| , \qquad \langle \mu - P\mu, g\rangle = \|\mu - P\mu\| .$$

The choices are possible because only finitely many point evaluations are involved.

Define

$$h(p) = f(\pi(p)) + g(p) - g(\pi(p)) \qquad (p \in M).$$

We claim that $h \in B_{\mathrm{Lip}_0(M)}$. If $\pi(p) = \pi(q)$, then

$$h(p) - h(q) = g(p) - g(q) \leq d(p, q).$$

If $\pi(p) \neq \pi(q)$, Lemma 4.1 gives

$$\begin{aligned} h(p) - h(q) &= g(p) - g(\pi(p)) + f(\pi(p)) - f(\pi(q)) + g(\pi(q)) - g(q) \\ &\leq d(p, \pi(p)) + d(\pi(p), \pi(q)) + d(\pi(q), q) = d(p, q). \end{aligned}$$

Interchanging p and q gives the reverse inequality, so h is 1-Lipschitz and $h(0) = 0$.

Pairing with μ gives

$$\|\mu\| \geq \langle \mu, h \rangle = \langle P\mu, f \rangle + \langle \mu - P\mu, g \rangle = \|P\mu\| + \|\mu - P\mu\| .$$

The opposite inequality is the triangle inequality. Hence P is an L-projection and $\mathcal{F}(M) = \mathcal{F}(A) \oplus_1 \ker P$. □

Proof of Corollary 2.2. Since A is a non-degenerate complete $\mathbb{R}$-tree in its own right, Godard's computation gives an isometric identification of $\mathcal{F}(A)$ with an L_1 space over the length measure of A. This measure is atomless. By the known L_1 characterization of Martin–Perreau–Rueda Zoca, every unit vector of this L_1 space is a super Daugavet-point. Equivalently, every unit vector of $\mathcal{F}(A)$ is a super Daugavet-point of $\mathcal{F}(A)$.

Theorem 2.1 gives

$$\mathcal{F}(M) = \mathcal{F}(A) \oplus_1 \ker P.$$

Super Daugavet-points pass from a summand to the ambient ℓ_1-sum, as in the stability result cited in the source paper from Martin–Perreau–Rueda Zoca. Therefore every $\mu \in S_{\mathcal{F}(A)}$ is a super Daugavet-point of $\mathcal{F}(M)$. □

5 Scope and limitations

This is a partial result, not a full solution of the source question. It handles every point lying in the free space of a closed metric-convex subtree of the ambient metric space. In particular it upgrades the source paper's single-segment L-summand mechanism to arbitrary closed subtrees and to all unit vectors in those subtrees.

The argument does not prove that an arbitrary Daugavet-point or Δ-point of $\mathcal{F}(M)$ must lie in such a subtree summand. That missing localization is precisely where the arbitrary-subset problem remains hard, especially when the point is represented through non-convex length-measure pieces or through the linear constraints caused by missing branch points.

Local index checks on June 29, 2026 found the previous run attempt on this arXiv id, which did not promote a result. The present packet is a new scoped positive theorem; it does not claim novelty for the background facts that free spaces over $\mathbb{R}$-trees are L_1 spaces or that L_1 Daugavet-points are super Daugavet-points.

6 References

[1] T. A. Abrahamsen, R. J. Aliaga, V. Lima, A. Martiny, Y. Perreau, A. Prochazka, and T. Veeorg, *A relative version of Daugavet-points and the Daugavet property*, arXiv:2306.05536, 2023.

[2] A. Godard, *Tree metrics and their Lipschitz-free spaces*, Proceedings of the American Mathematical Society 138 (2010), 4311–4320; arXiv:0904.3178.

[3] M. Martin, Y. Perreau, and A. Rueda Zoca, *Diametral notions for elements of the unit ball of a Banach space*, arXiv:2301.04433, 2023.